Historical Path-Residual Guided Candidate Re-ranking for Catalog-Assisted S-Phase Repicking

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Abstract

Automatic seismic phase picking with deep neural networks achieves high matched-timing accuracy, yet end-to-end errors remain dominated by S-phase wrong-peak selections on multi-modal probability curves. We propose a *catalog-assisted* S-phase candidate re-ranking method that leaves an external pretrained PhaseNet waveform model frozen and re-selects among its top-K=5 S candidates using a distance travel-time baseline, historical path residuals with Empirical-Bayes shrinkage (k=50), and a deterministic score mixing PhaseNet probability with history consistency. On an event-disjoint 10,000-trace Stage 3 chronological test set (INSTANCE), fixed re-ranking increases S-phase F1@0.5 from 0.695 to 0.718 and reduces end-to-end P95 absolute error from 2.15 s to 1.11 s relative to STEAD PhaseNet; event-level paired bootstrap confidence intervals exclude zero. A learned scalar gate does not outperform the fixed score and is reported only as an ablation. An auxiliary unused-event audit (9 events / 108 traces) shows the same direction across three external PhaseNet weight sets but is underpowered (bootstrap CIs include zero) and is not used as independent significance evidence. The method requires event/path context and falls back to PhaseNet when history is unavailable; it is not a blind continuous detector.

Contents

1 Introduction

Deep learning phase pickers such as PhaseNet [Zhu and Beroza, 2019] and Earthquake Transformer [Mousavi et al., 2020] have substantially improved automatic arrival-time estimation. When a pick is matched to a manual label within a tight window, residual timing errors are often only tens of milliseconds. In operational end-to-end evaluations, however, absolute-error distributions exhibit heavy tails: misses and wrong peaks—especially for S waves with multiple probability modes—dominate mean absolute error and high percentiles [Münchmeyer et al., 2022].

Most waveform pickers treat each trace independently and do not explicitly use historical travel-time residuals along repeated source-region–station paths. Catalog-assisted settings, where an approximate origin time and location are available (e.g., association after detection, or offline catalog refinement), provide precisely the context needed to score candidate peaks against a path prior without retraining the waveform network.

This paper addresses catalog-assisted *S-phase candidate re-picking / refinement*. Our contributions are:

1. We diagnose that end-to-end S-phase long tails on INSTANCE are driven by wrong-peak/miss behavior rather than matched micro-timing.
2. We introduce a deterministic re-ranking score combining a distance MLP travel-time baseline, median path residuals with shrinkage, and PhaseNet candidate probabilities.
3. On an event-disjoint Stage 3 test-only set of 10,000 traces we obtain statistically significant S F1@0.5 and e2e-P95 improvements versus STEAD PhaseNet under event-level paired bootstrap.
4. We document identity fallback without history, cross-weight transfer (STEAD/ETHZ/SCEDC), negative controls, and an underpowered unused-event Stage 4 audit; a learned gate does not beat the fixed score, so we do not escalate to GNNs.

We do *not* claim blind phase-picking SOTA, continuous detection capability, or that Stage 4 alone provides significant confirmation.

2 Related Work

Deep phase picking. PhaseNet [Zhu and Beroza, 2019] predicts sample-wise P/S probabilities with a U-Net-style architecture. EQTransformer [Mousavi et al., 2020] jointly detects events and picks phases. Generalized phase detection [Ross et al., 2018] and related CNN pickers similarly map waveforms to phase scores. Systematic benchmarks [Münchmeyer et al., 2022, Woollam et al., 2022] show that absolute performance depends strongly on dataset shift and evaluation protocol.

Travel times and path corrections. Classical seismology uses distance-dependent travel-time tables and station or path corrections estimated from residuals. Our distance MLP plays the role of a smooth baseline; historical median residuals supply a nonparametric path correction regularized by shrinkage.

Catalog-assisted and historical picking. We deliberately restrict scope to settings with event context. Methods that discover events from continuous waveforms (detection/association) are complementary and out of scope.

Candidate re-ranking. Peak finding on probability curves yields multiple S candidates; re-ranking with external constraints is a lightweight alternative to fine-tuning the picker. Our ablations show that a fixed score already captures most of the gain available from switching

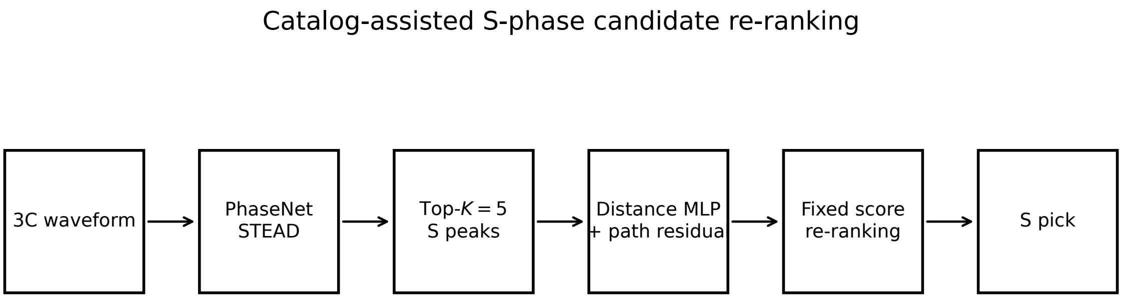

Figure 1: Catalog-assisted S-phase candidate re-ranking with a frozen PhaseNet.

between PhaseNet and a history-informed pick among existing candidates (oracle selector ceiling ≈ 0.720 F1@0.5), while oracle candidate selection (0.890) indicates remaining coverage limits.

Uncertainty-aware picking. Some pickers expose probability curves or calibrated scores; we treat PhaseNet probabilities as candidate features inside an explicit score rather than claiming a new uncertainty model. Additional uncertainty-calibration literature should be inserted only after manual DOI verification (placeholder avoided here).

3 Data and Evaluation Protocol

We use the INSTANCE Italian ML waveform dataset [Michelini et al., 2021]: 120 s windows at 100 Hz with three components (E,N,Z), accessed via SeisBench [Woollam et al., 2022]. Chronological train/val/test splits are event-disjoint. The external PhaseNet weight used for primary results is STEAD (not INSTANCE), to avoid training-data leakage from INSTANCE-pretrained weights; INSTANCE weights appear only in diagnostic experiments and are excluded from formal claims.

Evidence hierarchy.

- **Stage 3 (main test):** deterministic event-disjoint test-only list of 10,000 traces / 1,482 events.
- **Stage 2 (development):** mixed fixed-eval list containing validation and test events; used only for λ / shrinkage selection and exploratory analysis—never as an independent test set.
- **Stage 4 (auxiliary):** all remaining unused chronological test events after Stage 3 (9 events / 108 traces). Underpowered; directional audit only.

Metrics. We report Precision/Recall/F1 at ± 0.1 and ± 0.5 s, matched-timing errors on true positives, and end-to-end absolute errors on all labeled traces with a prediction (including wrong peaks). Miss and wrong-peak rates follow Stage 2 audit definitions. Significance uses event-level paired bootstrap (not i.i.d. traces).

4 Method

Figure 1 summarizes the pipeline. PhaseNet remains frozen; only candidate scores are adjusted.

4.1 PhaseNet candidates

Let $p_S(t)$ be the S-phase probability on the waveform grid after UTC remapping of SeisBench `annotate()`. We extract up to K=5 peaks with minimum distance 50 samples, prominence 0.05, and height 0.1; if none pass, the global argmax is kept as a fallback peak. Candidate j has sample index s_j, probability π_j, and prominence ρ_j.

4.2 Distance baseline and path residual

A distance MLP trained on chronological train predicts baseline travel times $\tau_{P/S}^{\text{base}}$. Path keys are *directed* source-region→station identifiers `region_i:region_j|depth_bin|station_id` with source region $\lfloor \text{lat}/\Delta \rfloor, \lfloor \text{lon}/\Delta \rfloor$ and grid Δ=0.2° (default), depth bins $\{0\text{–}5, 5\text{–}10, 10\text{–}20, 20\text{–}40, 40+\}$ km; keys are not undirected station pairs. For past *training* events on the matched path (with spatial/depth fallback levels in `TemporalHistoryStore`), residuals $r = \tau^{\text{obs}} - \tau^{\text{base}}$ are aggregated as median and MAD. Path residual use requires history count $n \geq 5$ and residual-P MAD ≤ 1.0 s (`mad_disable_s`); otherwise `history_available` is false and scoring falls back as below.

4.3 Shrinkage

With shrinkage parameter k=50 selected on Stage 2 train residual MAE,

$$w = \frac{n}{n+k}, \qquad \tilde{r} = w\, r^{\text{path}} + (1-w)\, r^{\text{global}}. \tag{1}$$

Predicted travel time is $\hat{\tau} = \tau^{\text{base}} + \tilde{r}$, converted to an expected sample $\hat{s}$ using origin time and trace start time. Uncertainty uses $\sigma = \text{clip}(1.4826 \cdot \text{MAD},\ 0.05,\ 1.0)$ s.

4.4 Fixed candidate score

History consistency is a Gaussian kernel in sample space,

$$h_j = \exp\Big(-\tfrac{1}{2}\big((s_j - \hat{s})/\sigma_{\text{samp}}\big)^2\Big). \tag{2}$$

The deterministic score (implementation in `candidate_rescorer.py`) is

$$\text{score}_j = \lambda_w \log(\pi_j + \varepsilon) + \lambda_h \log(h_j + \varepsilon) + \lambda_\rho \frac{\rho_j}{\max_{j'} \rho_{j'}}, \tag{3}$$

with frozen S-wave weights $(\lambda_w, \lambda_h, \lambda_\rho) = (0.5,\ 2.0,\ 0.0)$ chosen on Stage 2 validation (not Stage 3/4 test). If history is unavailable, λ_h is set to 0 (identity fallback toward probability ranking). The pick is $\arg\max_j \text{score}_j$. P picks remain PhaseNet for the S-focused Stage 3/4 comparisons.

4.5 Learned gate (ablation only)

A scalar gate mixing PhaseNet and history log-softmax rankings was trained in Stage 3 but did not beat fixed re-ranking on the test-only set and is not the recommended main method.

5 Experiments

All Stage 3/4 evaluations freeze PhaseNet STEAD inference, K=5, k=50, and Stage 2 validation λ's. Comparators include: PhaseNet; distance-only prior; raw path medians; MLP+residual without/with shrinkage; fixed catalog rescore (main); learned gate (3 seeds); shuffled/biased history controls; history-unavailable identity; oracle PN↔fixed selector; oracle nearest candidate

Table 1: Stage 3 event-disjoint test-only S-phase results (frozen STEAD PhaseNet).

Method	F1@0.1	F1@0.5	e2e P95 (s)	e2e MAE (s)
PhaseNet (STEAD)	0.462	0.695	2.15	1.99
fixed catalog_rescore	0.477	0.718	1.11	0.62
learned gate (ablation)	0.475	0.716	1.16	0.59
oracle PN/fixed selector	0.477	0.720	1.10	0.62
oracle candidate K=5	0.588	0.890	0.96	0.31

Table 2: Stage 3 event-level paired bootstrap: fixed re-ranking minus PhaseNet.

Metric	mean Δ	95% CI low	95% CI high
F1@0.5	0.0225	0.0186	0.0265
F1@0.1	0.0144	0.0113	0.0174
e2e P95 (s)	-1.0698	-1.7000	-0.6250
e2e MAE (s)	-1.3597	-1.6208	-1.1101
wrong-peak rate	-0.0275	-0.0323	-0.0227

in top-K. Cross-weight transfer applies the same frozen rescore to ETHZ and SCEDC PhaseNet outputs on the Stage 4 holdout. Event-level paired bootstrap uses 2000 replicates on Stage 3 and 5000 on Stage 4. Hard subsets use Stage 2 predefined bins (no holdout retuning).

6 Results

6.1 Main test (Stage 3)

Table 1 reports Stage 3 test-only S-phase metrics (10,000 traces / 1,482 events). Fixed re-ranking improves F1@0.5 from 0.695 to 0.718 and reduces e2e P95 from 2.15 s to 1.11 s ($\approx$ 48% relative P95 reduction; absolute values retained). F1@0.1 also rises (0.462 $\rightarrow$ 0.477), but the larger operational gain is in suppressing gross wrong-peak tails rather than claiming sub-sample-point timing refinement.

Event-level bootstrap of fixed versus PhaseNet (frozen predictions, seed 42, n=2000) yields ΔF1@0.5=0.0225 with 95% CI [0.019, 0.026] and ΔP95 with CI entirely below zero (Table 2; `bootstrap_fixed_vs_phasenet.json`). Learned gate versus fixed has ΔF1@0.5 CI [-0.004, -0.0004] (gate slightly worse).

Figure 2 shows error CDFs; fixed dominates the long tail. Figure 3 reports F1 at 0.1 s and 0.5 s. Figure 4 shows Stage 2 development hard-subset gains (motivational; not a test claim). Figures 5–6 illustrate a wrong-peak repair and a failure mode from frozen Stage 3 case dumps.

6.2 Auxiliary unused-event audit (Stage 4)

On the nine previously unused test events (108 traces), fixed re-ranking moves S F1@0.5 from 0.678 to 0.690 and P95 from 1.03 s to 0.73 s, with the same direction for ETHZ/SCEDC weights and exact PhaseNet match under history-unavailable scoring. However, event-level bootstrap CIs include zero; the audit is underpowered and is reported only as a directional unused-event check (*directionally replicated in an underpowered unused-event audit*), not as independent significance validation.

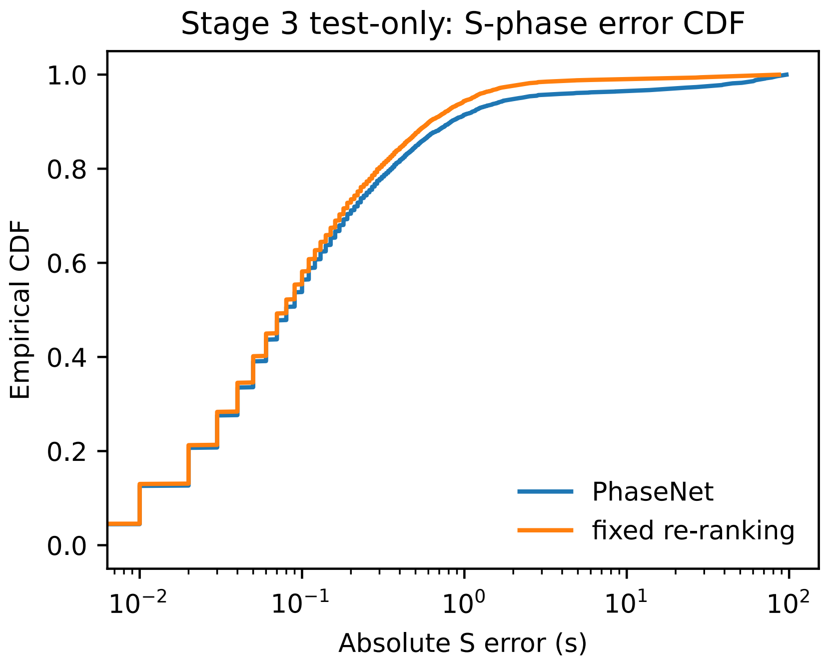

Figure 2: Stage 3 S-phase absolute-error CDF: PhaseNet vs fixed re-ranking.

6.3 Ablations and ceilings

Oracle candidate-K=5 reaches F1@0.5=0.890, while oracle PN/fixed selector only 0.720, indicating that (i) candidate coverage still limits performance and (ii) a binary gate between PhaseNet and fixed has little headroom—consistent with the learned gate collapsing toward PhaseNet. Shuffled/biased history controls degrade relative to correct history on development and Stage 4 audits.

7 Discussion

The Stage 3 gain pattern—clearer at F1@0.5 and e2e P95 than at ultra-tight matched timing—supports interpreting the method as wrong-peak repair under catalog constraints. The high oracle-candidate ceiling shows many labels already lie near some PhaseNet peak; re-ranking exploits that coverage. The low oracle-selector ceiling and near-constant learned gate explain why a simple fixed score is preferable to a learned switch and why we stop before GNNs: additional model capacity did not beat a transparent deterministic rule on the main test set.

Catalog assistance is both a strength (usable path priors) and a boundary: without event context the method must identity-fallback and cannot replace continuous detection/association.

8 Limitations

- Requires catalog/event context (origin time and path features); not a blind continuous detector or associator.
- Depends on historical path coverage; sparse paths shrink toward the distance baseline / PhaseNet.
- Source-region path keys approximate spatial structure.

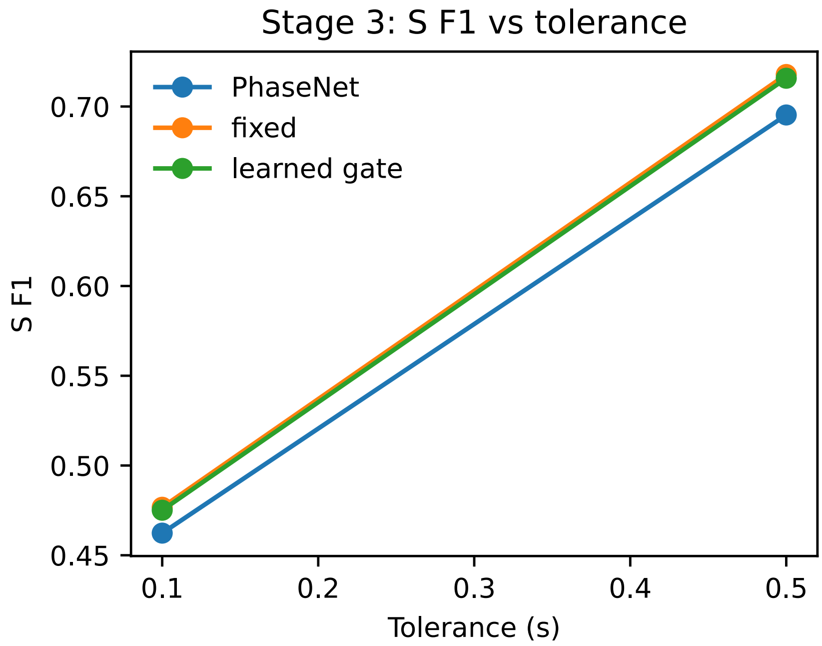

Figure 3: Stage 3 S F1 versus tolerance.

- Stage 4 retains only nine unused test events after Stage 3 consumption; it cannot alone support significant claims.
- Primary evidence is INSTANCE; we do not claim cross-dataset SOTA.
- Improvements target S-phase gross errors; P remains PhaseNet in the reported S-focused protocol.
- Catalog origin-time and location errors can bias expected arrivals.
- λ and k were selected on Stage 2 development validation; Stage 3 test metrics were inspected when choosing fixed over gate as the main method, which we disclose.

9 Conclusion

We presented a catalog-assisted, history-residual-guided re-ranking of PhaseNet S-wave candidates that needs no picker fine-tuning. On the Stage 3 event-disjoint 10k test, fixed re-ranking significantly improves S F1@0.5 and reduces e2e P95 versus STEAD PhaseNet; a learned gate does not outperform it. Stage 4 provides only an underpowered directional audit. We recommend the fixed score as the main method and advise against claiming blind phase-picking SOTA or escalating to GNN complexity on this evidence.

References

Alberto Michelini, Spina Cianetti, Sonja Gaviano, Carlo Giunchi, Dario Jozinović, and Valentino Lauciani. INSTANCE – the Italian seismic dataset for machine learning. *Earth System Science Data*, 13(12):5509–5544, 2021. doi: 10.5194/essd-13-5509-2021.

S. Mostafa Mousavi, William L. Ellsworth, Weiqiang Zhu, Lindsay Y. Chuang, and Gregory C. Beroza. Earthquake transformer—an attentive deep-learning model for simultane-

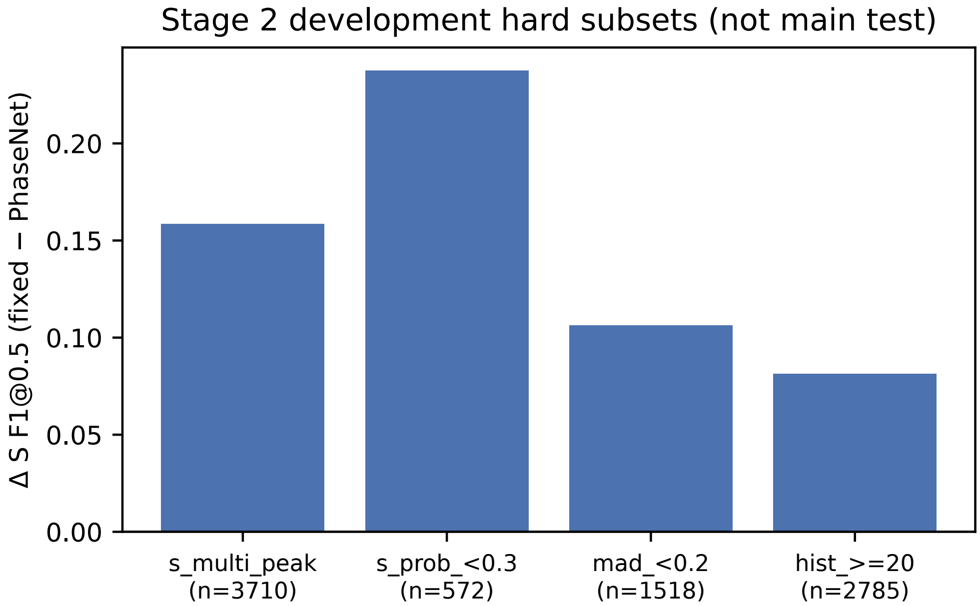

Figure 4: Stage 2 development hard-subset ΔF1@0.5 (mixed fixed-eval; not main test).

ous earthquake detection and phase picking. *Nature Communications*, 11:3952, 2020. doi: 10.1038/s41467-020-17591-w.

Jannes Münchmeyer, Jack Woollam, Andreas Rietbrock, Frederik Tilmann, Dietrich Lange, Thomas Bornstein, Tobias Diehl, Carlo Giunchi, Florian Haslinger, Dario Jozinović, Alberto Michelini, Joachim Saul, and Hugo Soto. Which picker fits my data? A quantitative evaluation of deep learning based seismic pickers. *Journal of Geophysical Research: Solid Earth*, 127(1): e2021JB023499, 2022. doi: 10.1029/2021JB023499.

Zachary E. Ross, Men-Andrin Meier, Egill Hauksson, and Thomas H. Heaton. Generalized seismic phase detection with deep learning. *Bulletin of the Seismological Society of America*, 108(5A):2894–2901, 2018. doi: 10.1785/0120180080.

Jack Woollam, Jannes Münchmeyer, Frederik Tilmann, Andreas Rietbrock, Dietrich Lange, Thomas Bornstein, Tobias Diehl, Carlo Giunchi, Florian Haslinger, Dario Jozinović, Alberto Michelini, Joachim Saul, and Hugo Soto. SeisBench—A toolbox for machine learning in seismology. *Seismological Research Letters*, 93(3):1695–1709, 2022. doi: 10.1785/0220210324.

Weiqiang Zhu and Gregory C Beroza. PhaseNet: a deep-neural-network-based seismic arrival-time picking method. *Geophysical Journal International*, 216(1):261–273, 2019. doi: 10.1093/gji/ggy423.

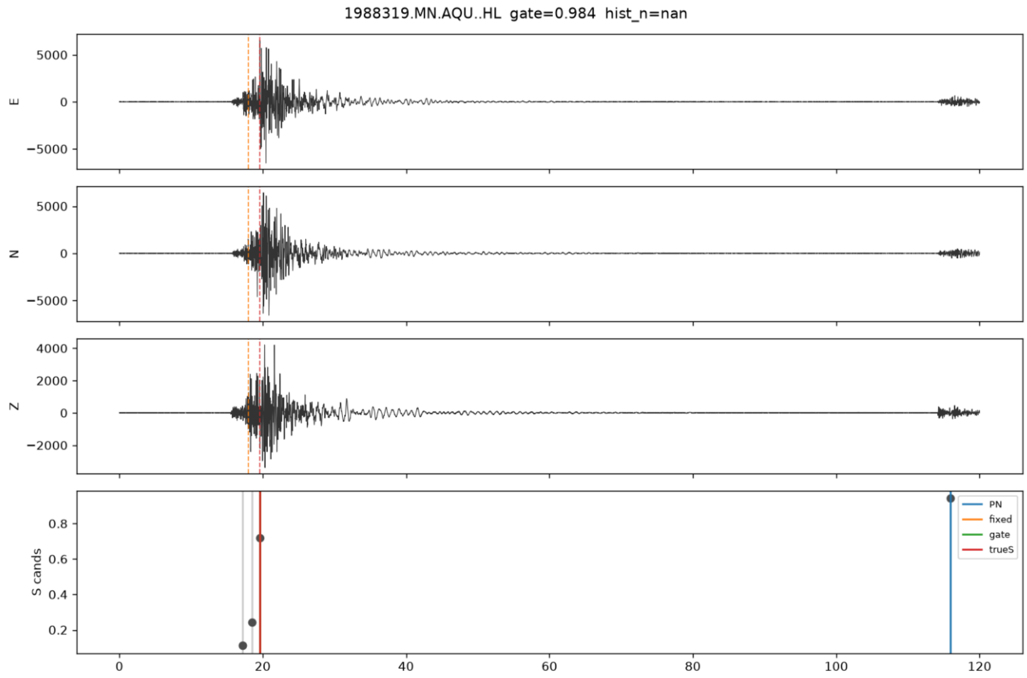

Figure 5: Qualitative success: PhaseNet wrong peak corrected by fixed re-ranking (frozen case dump).

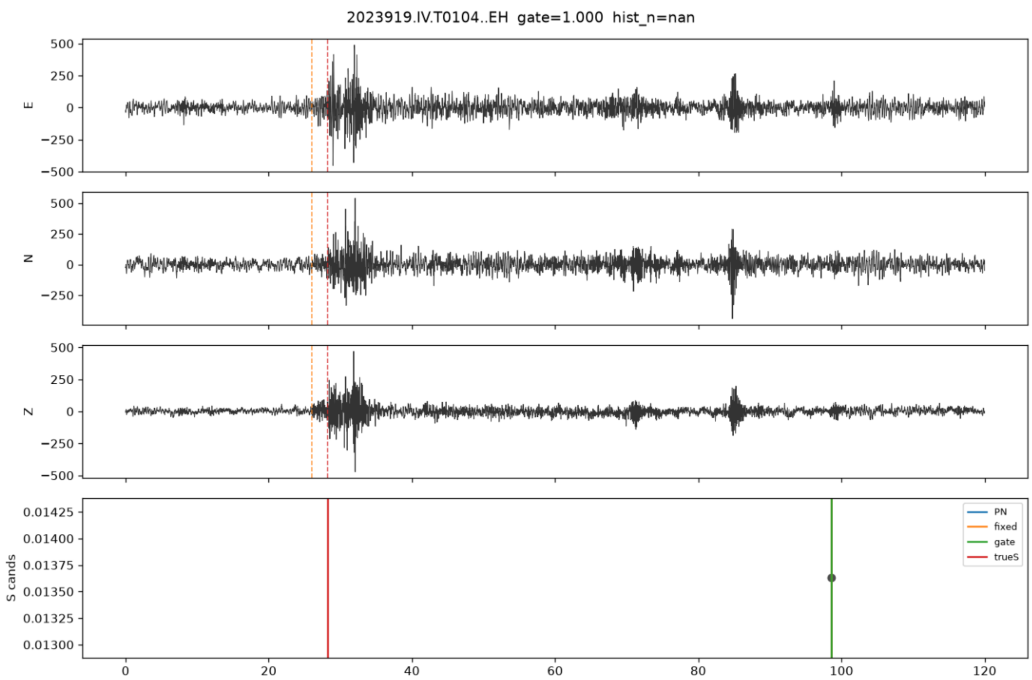

Figure 6: Qualitative failure: no correct candidate in top-K (oracle-candidate gap).